

How the pieces fit

Agent · LLM · Skill · MCP · Tools · Schema

One data flow, built one piece at a time. Advance with the arrow key.

A language model, by itself

1 / 9



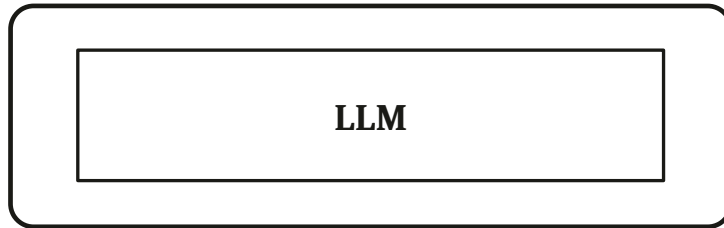
text $\rightarrow$ a distribution over next tokens

Text in, a distribution over next tokens out. Stateless: nothing carries between calls, and the weights never change.

Put it in a loop — that is an agent

2 / 9

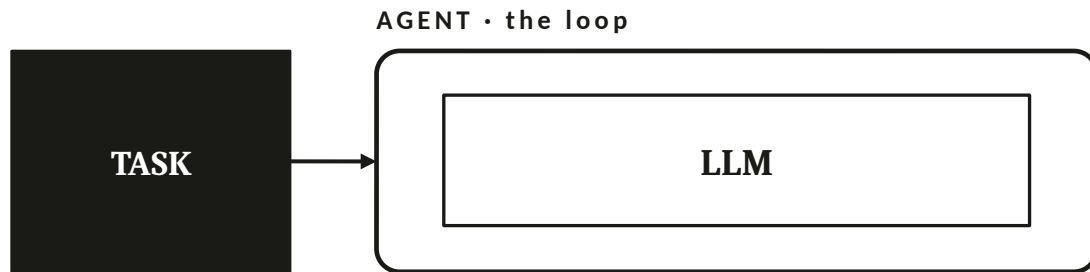
AGENT • the loop



The agent calls the model repeatedly, decides what to do next, and stops when it judges the work done. The model is the engine; the loop is the agent.

Give the loop a task

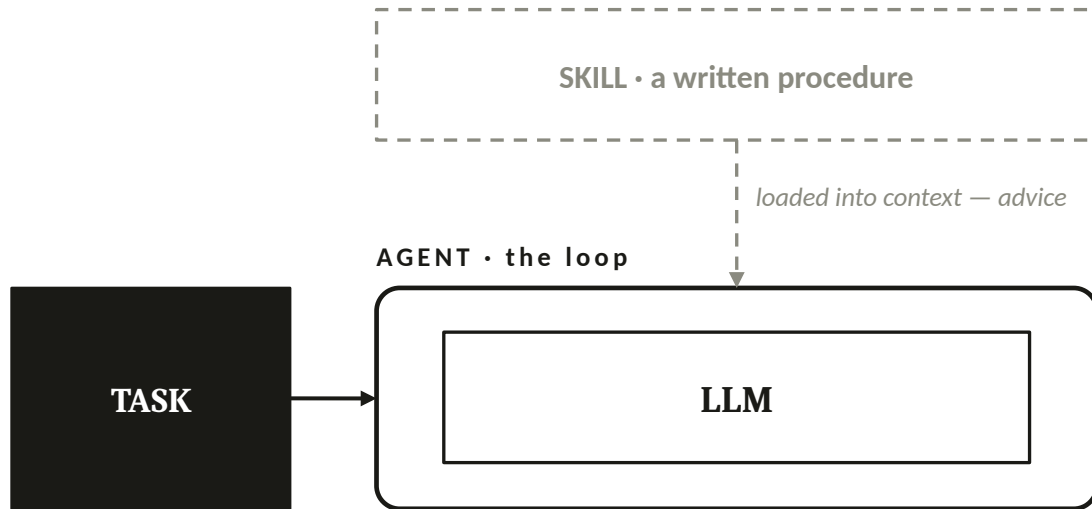
3 / 9



A bounded work order arrives: what to do, on which inputs, and what counts as finished.

A skill is a written procedure

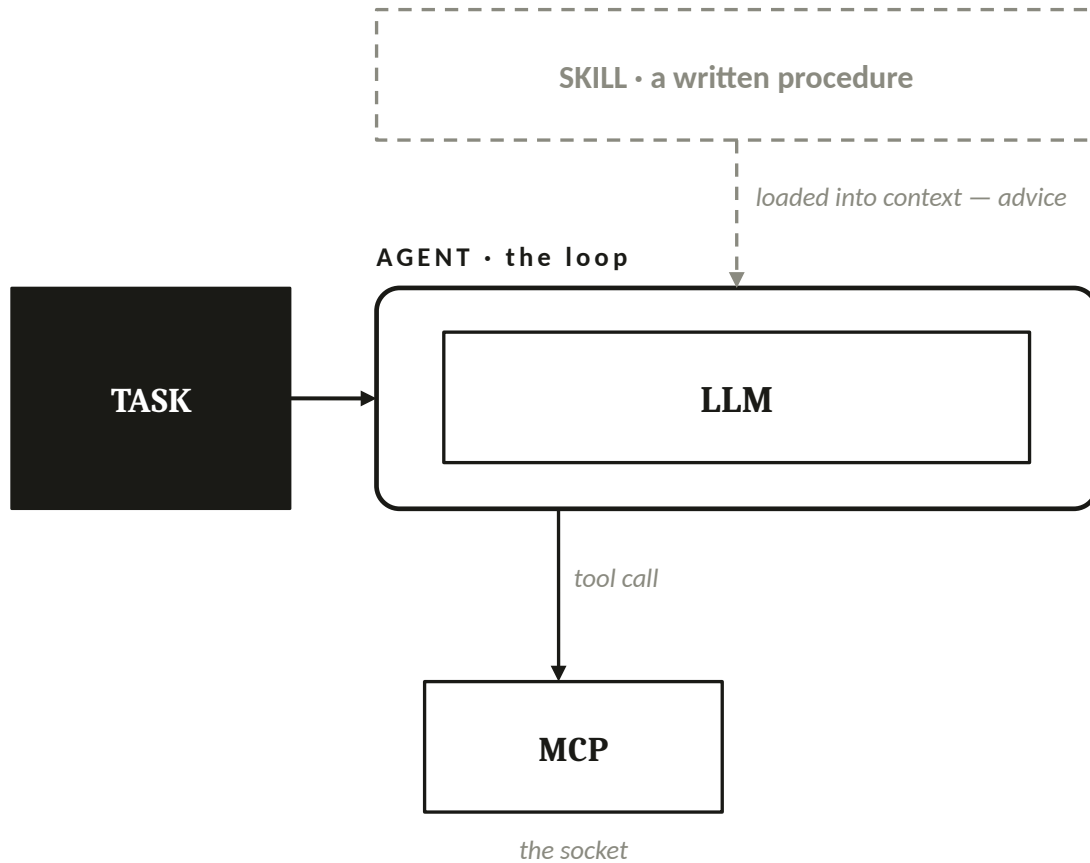
4 / 9



A protocol loaded into the context when the task matches it — drawn dashed, because it is advice. The model can read it and not follow it, and nothing records that it did.

MCP is the socket, not the tool

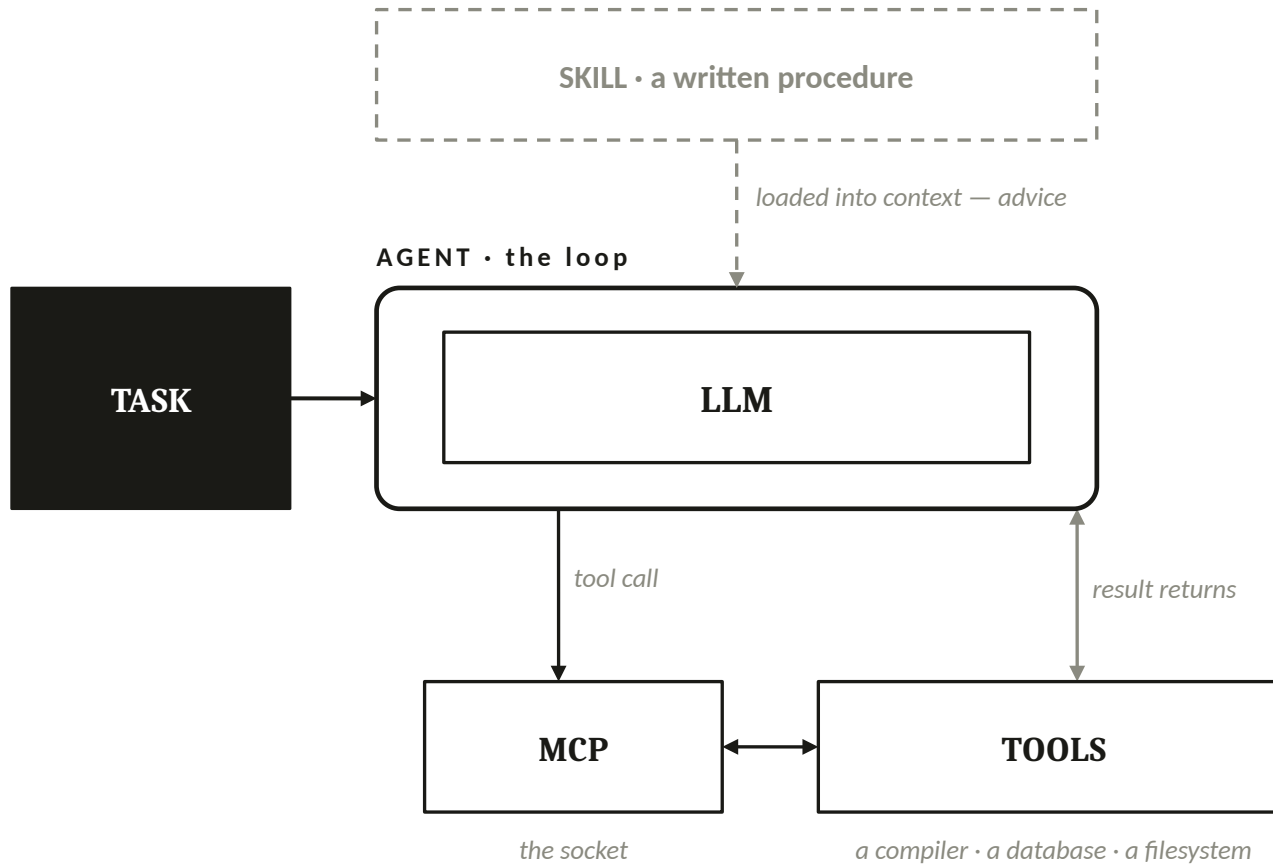
5 / 9



A published convention for exposing a tool so that any model can call it. It standardises the plug; it says nothing about what comes through it.

Tools are what actually act

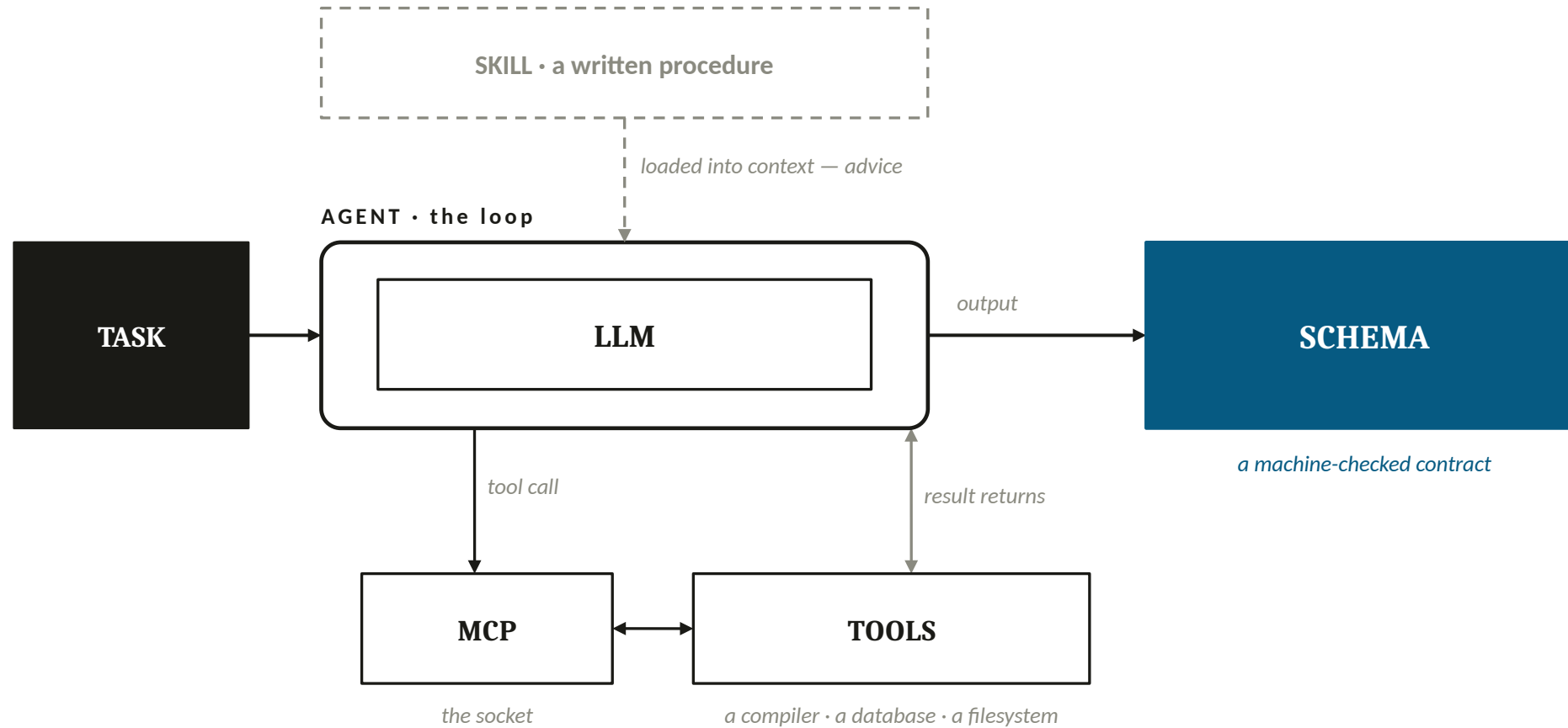
6 / 9



A compiler, a database, a filesystem. The result returns to the loop, which calls the model again — this return arrow is the whole reason it is called an agent.

The output meets a schema

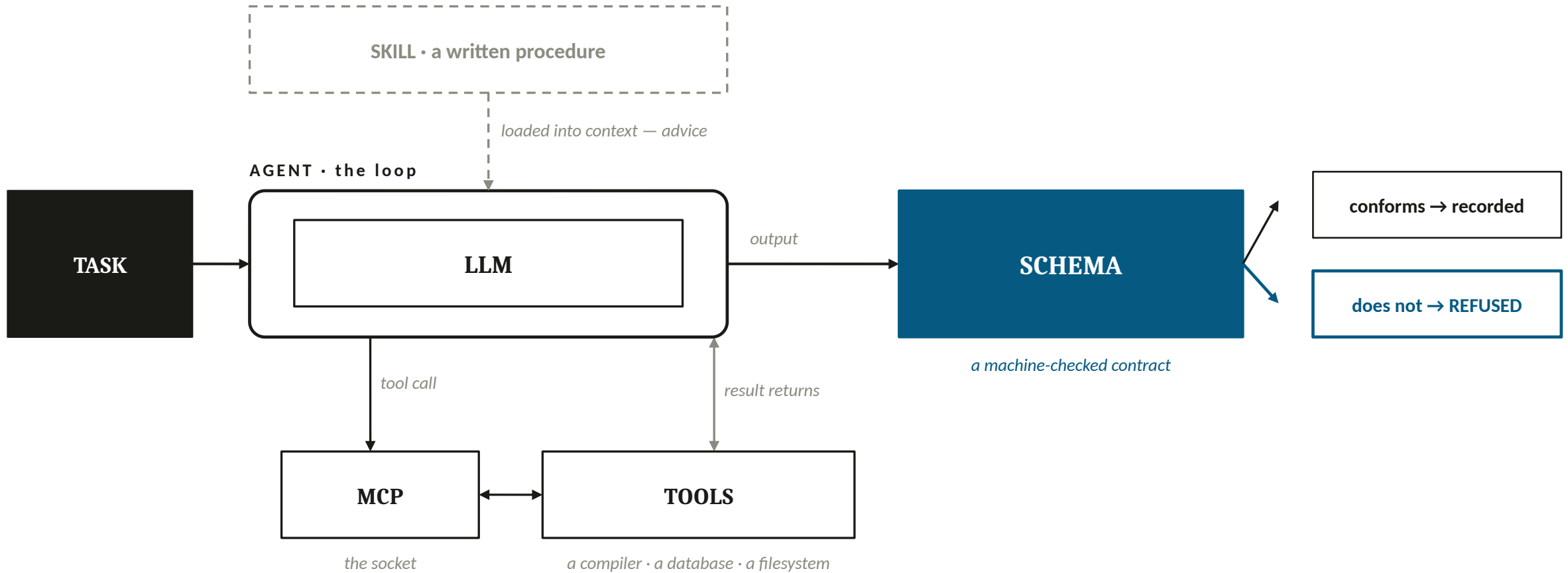
7 / 9



A machine-checked contract: which fields a valid record must have, which types, and — crucially — that no field nobody agreed to may be added.

And the schema can say no

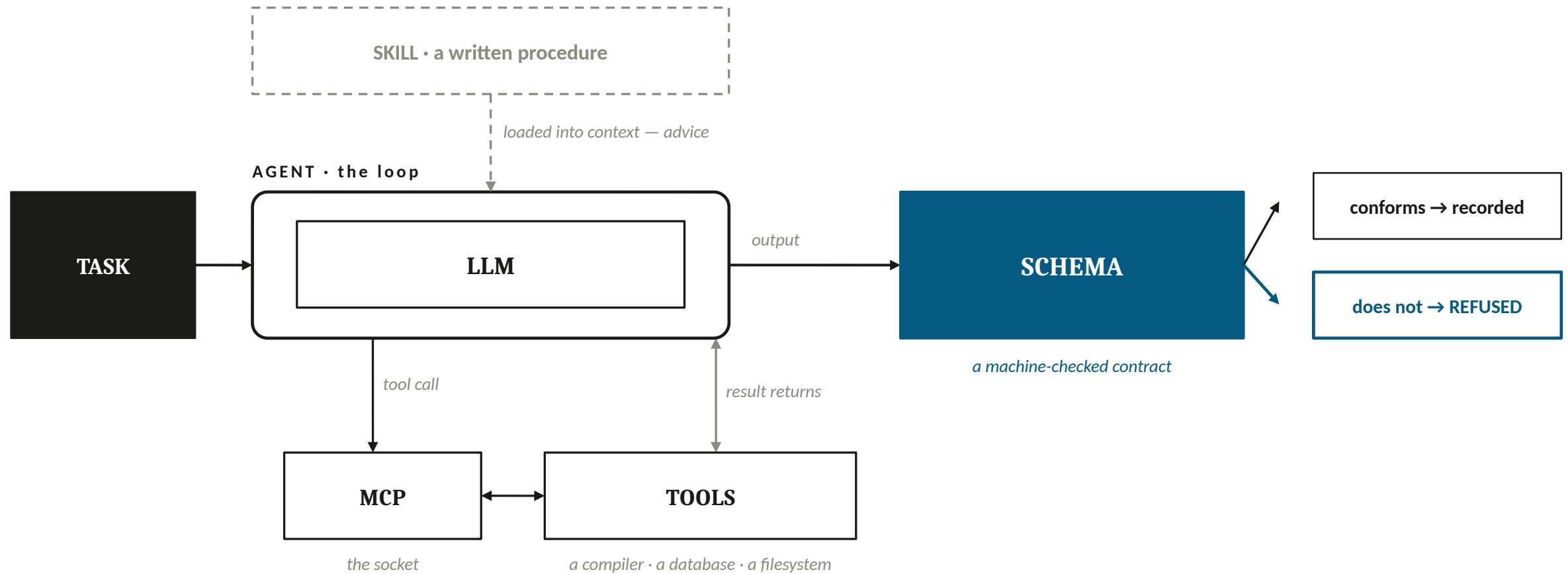
8 / 9



This is the only arrow on the diagram that rejects. A record that does not conform is not repaired or accepted with a warning; it is refused.

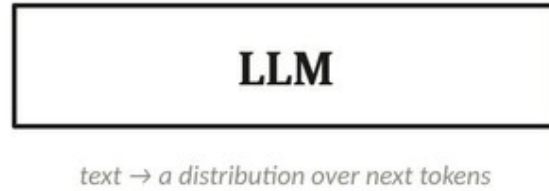
Three ways of asking. One way of refusing.

9 / 9



The prompt, the skill and the tool call are all requests — they can be ignored, and the transcript looks the same either way. Only the schema can make an answer impossible to express.

The whole flow, looping



Plays automatically in slideshow mode (press F5 / Play). The nine slides before this one are the same build, under your own clicks — use whichever suits the room.